QUANT

Work & Interest

Assignment 03

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Work, Chain Rule & Interest Online Assignment -03

1. A can do some work in 24 hours, B can do the same work in 32 hours and C can do the same work in 60 hours. They start working together. A left after 6 hours and B left after working for 8 hours. How many more hours are required to complete the whole work after B has left?

1. 30 hours 2. 25 hours 3. 17 hours 4. 22 hours 5. 20 hours

2. A highly specialized work can be done by Mr. Laxmipati Balaji, Mr. Ashish Nehra and Mr. Zaheer Khan in 12 days. The same work can be done by Mr. Ashish Nehra, Mr. Zaheer Khan and Mr. Irfan Pathan in 15 days. The same very work can be done by Mr. Laxmipati Balaji and Mr. Irfan Pathan in 20 days. How much time it will take if they all work together?

1. 5 days 2. 12 days 3. 10 days 4. 8 days 5. 6 days

3. In the beginning Ram works at a rate such that he can finish a work in 12 hrs, but he only works at this rate for 8 hrs. After that he works at a rate such that he can do the whole work in 21 hrs. If Ram is to finish this work at a stretch, how many hours will he take to finish this work?

1. 12 hrs 2. 18 hrs 3. 11½ hrs 4. 15 hrs 5. 20 hrs

4. A leak in the bottom of a tank can empty the full tank in 6 hours. An inlet pipe fills water at the rate of 5 litres per minute. When the tank is full, the inlet is opened and due to the leak the tank is emptied in 8 hrs. What is the capacity of the tank (in litres)?

1. 5400 2. 1800 3. 14400 4. 3600 5. 7200

5. Chanda is twice as efficient as Mama and together they can finish a piece of work in 20 days. Mama will take how many days to finish this work alone?

1. 30 days 2. 60 days 3. 15 days 4. 12 days 5. 10 days

6. X can do a work in 60 days. Y can do this work in 40 days. They finish the work in 20 days with the help of Z. If a total of Rs. 240 were received for the whole work, how much out of this should be paid to Z?

1. Rs. 80 2. Rs. 120 3. Rs. 160 4. Rs. 40 5. Rs. 25

7. Pipe A and B running together can fill a cistern in 12 minutes. If B takes 10 minutes more than A to fill the cistern, then the time in which A and B will fill half the cistern separately will be respectively (in minutes)?

1. 30, 20 2. 15, 10 3. 10, 15 4. 20, 30 5. 15, 20

8. A, B, C and D can do a piece of work in 40 days. If A and B can do it together in 100 days, and C alone in 120 days, find the time in which D alone can do it.

1. 120 days 2. 200 days 3. 150 days 4. 90 days 5. 60 days

9. M can fill a tank in 20 minutes. N can fill this tank in 30 minutes and O can fill this tank in 40 minutes. All three are opened together. M is closed after 5 minutes. After how many minutes from the beginning should O be closed so that the tank gets filled in 15 minutes?

1. 10 min. 2. 15 min. 3. 5 min. 4. 6 min 5. 3 min

10. A bathtub can be filled by the cold water pipe in 10 minutes and by a hot water pipe in 15 minutes. Mr. Gokulmathura leaves the bathtub after turning on both the pipes simultaneously and returns at the moment, when the bathroom should be full. However, after finding that the waste pipe has been open, he now closes it. In 4 more minutes the bathtub is full. In what time can the waste pipe empty the full bathtub, working alone?

1. 9 min. 2. 12 min. 3. 18 min. 4. 12 min. 5. 10 min

Answer Key & Explanations

1.	4	In 6 hours *A* would have done 6/24 = 1/4th of the work. In 8 hours *B* would have done 8/32 = 1/4th of the work. *A* and *B* would have done ¼ + ¼ = ½ of the work in 8 hours. The remaining work in ½, which *C* has done. C can do half the work in 60 × ½ = 30 hours. Now out of these 30 hours *C* has already spent 8 hours along with *A* and *B*. Thus it will take 30 – 8 = 22 hours more to finish the work.
2.	3	The question specifies the time taken by 1st, 2nd and 3rd person. Then by 2nd, 3rd and 4th person. Then the time taken by 1st and 4th person. Now adding all three gives you the time taken by 2 times 1st, 2nd, 3rd and 4th person. Thus the time taken by 2 times all the persons will be 1/12 +1/15 + 1/20 = 12/60 = 1/5th = 5 days. Thus the four persons together will take double that time i.e. 5 × 2 = 10 days.
3.	4	Ram's 8 hr work = 8/12 = 2/3. Remaining work = 1 – 2/3 = 1/3. This will be completed in 1/3 × 21 i.e. 7 hrs. So total time taken to complete work = 8 + 7 = 15 hrs.
4.	5	Leak can empty in 6 hrs. Rate of water inlet = 5 liters per minute. Now the tank will be emptied in 8 hrs. ∴ Inlet pipe will take 1/6 – 1/8 i.e. 1/24 i.e. 24 hrs. to fill the tank. Hence capacity of the tank = 24 × 60 × 5 = 7200 liters.
5.	2	Chanda is double efficient than Mama. Let Mama takes 2*x* days and ∴ Chanda takes *x* days to complete the work. ∴ 1/*x* + 1/2*x* = 1/20 ⇒ *x* = 30. ∴ Mama will take 30 × 2 = 60 days to complete the work.
6.	4	Amount of work done by *X* = 20/60 = 1/3. Amount of work done by *Y* = 20/40 = ½. ∴Work done by *Z* = 1 – 1/3 – ½ = 1/6. ∴Amount paid to *Z* = 1/6 × 240 = Rs. 40.
7.	3	The two equation will be made as 1/*A* + 1/*B* = 1/12 and *A* + 10 = *B*. Solving these two you get *A* and *B* to be 20 and 30 respectively. But the question is asking the time they will take to fill half the tank, thus the ans will be 10 and 15 i.e. 3rd opt.
8.	3	*D* alone will take 1/40 – 1/100 – 1/120 = 1/150 ⇒ 150 days to complete the work.
9.	1	M's work for 5 min. = 5/20 = ¼. *N* worked for 15 min. ∴ *N*'s work for 15 min = 15/30 = ½. Remaining work i.e.1 – ½ – ¼ = ¼ is done by *O*. So *O* completed ¼ of the work in ¼ × 40 i.e. in 10 minutes.
10.	1	Working together, bathroom should be full in 1/10 + 1/15 i.e. 1/6 ⇒ 6 min. Hence waste pipe has worked for 6 min. This means that work done by these 2 filling pipes in 4 min is the same as the work done by waste pipe in 6 min. Work done by filling pipes in 4 min. = 4/6 = 2/3. ∴ In 6 min, waste pipe completes 2/3 of the work, hence total work done by the waste pipe will take 6 × 3/2 i.e. 9 min.